

SAQLAIN SHAH

JUNIOR ML & DL ENGINEER

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SUMMARY

Final-year BS Computer Science student at the University of Lahore and aspiring ML & DL Engineer with hands-on experience in Machine Learning, Deep Learning, and AI system development. Skilled in building end-to-end real-world AI projects using Python — including computer vision, automation systems, SAR satellite image analysis, and LSTM-based time series forecasting. Strong understanding of neural networks, sequence models, and Flask-based model deployment. Currently advancing into Generative AI by actively studying Transformers, LangChain, and LangGraph to transition into LLM-based application development. Seeking to apply and expand technical expertise in a professional AI/ML engineering environment.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Assembly Language
 - **Machine Learning:** Regression, Classification, Clustering (K-Means), Model Evaluation, Time Series Forecasting
 - **Deep Learning:** CNN, RNN, LSTM, Sequence Modeling, SAR Image Segmentation
 - **Libraries & Tools:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras, OpenCV, CVZone, Flask, Matplotlib, yFinance
 - **Other:** REST APIs, Data Preprocessing, Model Deployment, Git, Linux
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EXPERIENCE

AI/ML Junior Engineer Intern — Nolyth

- Worked across Machine Learning, Deep Learning, and Generative AI tasks, with a strong focus on backend development
 - Collaborated on fullstack projects, building backend services and integrating them with frontend interfaces
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PROJECTS

Sentinel Flux Core: Modular SAR Flood Detection Toolkit (Final Year Project)

- Developed AI flood detection system using Sentinel-1 SAR satellite imagery across multiple Pakistan regions
- Achieved Macro F1-score of 0.874 using hybrid ML + rule-based models with calibration metrics (Brier Score, ECE)
- Built impact analysis linking flood zones with population, roads, and infrastructure data
- Delivered reproducible pipeline with preprocessing, evaluation, and dashboard integration

TripMind AI: Multi-Agent Travel Planner

- Built a multi-agent AI travel planning app in Python and Streamlit, generating personalized itineraries, budgets, and weather-aware plans
- Switched from CrewAI to direct Groq API calls after hitting version incompatibilities, keeping the multi-agent architecture intact
- Deployed and tested locally via Streamlit on Arch Linux

Stock Price Prediction Web App (LSTM + Flask)

- Built a deep LSTM model on 20+ years of AAPL data via yFinance API for accurate stock price forecasting
- Engineered 100-day lookback window with MinMaxScaler normalization to capture long-term price trends
- Resolved critical inverse transform data leakage bug ensuring real-world accurate price predictions
- Deployed Flask app with MA, EMA, and Actual vs Predicted price visualizations
- Persisted trained model via model.save() for real-time inference without retraining

Virtual Keyboard with Hand Tracking

- Developed real-time gesture-based typing system using OpenCV and CVZone
- Implemented hover-based key press detection and shortcut buttons for enhanced user interaction

Dog vs. Cat Image Classification

- Built CNN-based image classification model using TensorFlow and Keras
- Applied data preprocessing and model evaluation techniques to achieve high classification accuracy

Voice-Controlled Desktop Assistant

- Created Python-based desktop assistant using Speech Recognition and Pyttsx3
- Programmed voice commands to automate system tasks, fetch news, and open websites

Face Recognition Attendance System

- Automated attendance logging using face_recognition and OpenCV libraries
- Recorded real-time attendance with CSV timestamps for accurate record keeping

EDUCATION

University of Lahore

2023 – Present

Bachelor of Computer Science | GPA: 3.54

AWARDS & CERTIFICATIONS

- Machine Learning Specialization – Coursera
- Deep Learning Specialization – Coursera
- Python Specialization – Coursera
- Improving Deep Neural Networks: Hyperparameter Tuning – DeepLearning.AI
- Neural Networks and Deep Learning – DeepLearning.AI